

OPHIRA

Reflectometry-fitting toolkit for porous silicon and thin films

D2 - Physics & Math Reference

Detailed reference on the physics of the forward model (Fresnel/Airy, anisotropy, angular spread), on the fit objective (chi-square, regularization, monotonicity, anti-zigzag), on the uncertainty estimation (diagonal Hessian, KK propagation), and on the adopted Kramers-Kronig relation (MSKK-1). For each formula the physical meaning is given and, where relevant, the **why** of a specific numerical choice in the code.

Related documents: D1 (user manual), D3 (operating procedures). The code pointers here are kept generic (function or module name).

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Conventions	wavelengths in nm, energies in eV, angles in degrees, complex indices $n + i k$

Contents

D2 - Physics & Math Reference	1
Contents	2
1 Introduction	4
1.1 Notation and symbols	4
2 Forward model	5
2.1 Complex Snell's law	5
2.2 Fresnel coefficients	5
2.3 Airy summation	5
2.4 Polarization collapse and unpolarized combination	5
2.5 Empirical fringe-contrast envelope	6
2.6 Effective-index difference	6
2.7 Angular spread (HOBL configuration)	6
3 $n(\lambda)$ and $k(\lambda)$ model	8
3.1 Base nodes and extended nodes	8
3.2 PCHIP interpolation	8
3.3 Nonlinear slider maps	8
4 Instrumental corrections	9
4.1 Roughness and scattering: f_{scatt}	9
4.2 Lateral inhomogeneity: f_{amp}	9
4.3 Global Debye-Waller: f_{DW}	9
4.4 IR correction at the detector change ($\lambda > 890$ nm)	10
4.5 Measuring the IR jump from the data (MEAS JUMP)	10
4.6 Additive offset	10
5 Fit objective function	11
5.1 Main term: weighted residuals	11
5.2 Optical-thickness anchoring term (HOBL only)	11
5.3 Regularization (smoothness)	11
5.4 Monotonicity ($\lambda \geq 420$ nm, for both n and k)	12
5.5 Convexity of n and k	12
5.6 Anti-zigzag (both n and k)	12
5.7 IR maxima position anchoring (40 deg and 60 deg)	12

5.8	Optional maxima-centering aid (FRINGE AID)	13
5.9	Total functional	13
6	Optimization	14
6.1	L-BFGS-B algorithm	14
6.2	Lock system	14
6.3	Multi-stage in run_fit_multi	14
7	Uncertainty estimation	15
7.1	Numerical diagonal Hessian	15
7.2	Limitations	15
7.3	Propagation sigma -> sigma_n_KK via Hilbert	15
8	Kramers-Kronig relation	16
8.1	Adopted Kramers-Kronig form: MSKK-1	16
8.2	Cauchy singularity	16
8.3	Boundary artifact at 2550 nm	16
8.4	Savitzky-Golay smoothing	16
8.5	Consistency metric	16
9	Thickness triangulation (Bragg/Snell)	18
9.1	Bragg condition for the maxima	18
9.2	Dispersion-free two-angle solution	18
9.3	Effective index from single-angle maxima (n_{eff})	18
9.4	Adaptive maxima detection	19
9.5	Determining d as a bracket: RMS residual(d) and KK residual(d)	19
10	Physical constraints and priors	20
10.1	Monotonicity of n only at $\lambda \geq 420$ nm	20
10.2	k approximately 0 in the transparency region (K-BOUND)	20
10.3	KK valid only at 8 deg	20
11	Symbol glossary	21

1 Introduction

This document summarizes the mathematics and physics of the OPHIRA code in its current form. The purpose is not to replace the source code, but to provide a self-contained reference to the model and the methods it implements.

The program extracts $n(\lambda)$ and $k(\lambda)$ of a thin film from reflectance spectra at multiple incidence angles, using a **data-first, physics-as-validation** approach: no dielectric model imposed a priori, but physical constraints of smoothness, monotonicity and optical-thickness anchoring, plus subsequent verification via Kramers-Kronig and geometric triangulation.

1.1 Notation and symbols

Notation used throughout the document:

- λ : vacuum wavelength (nm).
- $E = 1240/\lambda$: energy in eV.
- $n + i k$: complex refractive index of the film. n and k real, $k \geq 0$ for absorption.
- $\epsilon_1 + i \epsilon_2 = (n + i k)^2$: dielectric function.
- θ_0 : incidence angle in air (degrees in the code, radians in the formula).
- θ_1 : refraction angle in the film (complex, from Snell).
- d : physical film thickness (nm).
- $R(\lambda)$: reflectance, in % ($0 \leq R \leq 100$).
- s, p : TE (s) and TM (p) polarizations.
- $pol \equiv pol \in [0, 1]$: s-polarized intensity fraction of the source.

2 Forward model

The simulated reflectance $R(\lambda)$ is computed in `physics.calculate_refl_core(...)`. The geometry is: air (medium 0) - homogeneous film of thickness d and index $n_f + i k_f$ (medium 1) - semi-infinite substrate of index $n_{sub} + i k_{sub}$ (medium 2). The substrate is treated as semi-infinite because the wafer backside is typically unpolished and scatters the IR out of the specular cone.

2.1 Complex Snell's law

For complex indices the refraction angles are complex. Sine conservation:

$$\sin \theta_1 = \frac{\sin \theta_0}{n_f + i k_f}, \quad \cos \theta_1 = \sqrt{1 - \sin^2 \theta_1}$$

The real part of $\cos \theta_1$ determines the propagating component; the imaginary part controls the evanescent decay. An analogous formula holds for θ_2 in the substrate. The code uses the convention `cos1 = np.sqrt(1 - sin1**2 + 0j)` to guarantee the positive branch of the exponent.

2.2 Fresnel coefficients

For each interface, the Fresnel coefficients for the reflected electric fields are (sign convention: incident field, r_s with $v \perp$ plane of incidence, r_p with v in the plane):

$$r_s^{01} = \frac{\cos \theta_0 - n_f \cos \theta_1}{\cos \theta_0 + n_f \cos \theta_1}$$

$$r_p^{01} = \frac{n_f \cos \theta_0 - \cos \theta_1}{n_f \cos \theta_0 + \cos \theta_1}$$

$$r_s^{12} = \frac{n_f \cos \theta_1 - n_{sub} \cos \theta_2}{n_f \cos \theta_1 + n_{sub} \cos \theta_2}$$

$$r_p^{12} = \frac{n_{sub} \cos \theta_1 - n_f \cos \theta_2}{n_{sub} \cos \theta_1 + n_f \cos \theta_2}$$

In the current (collapsed) model the film is treated as isotropic, so r_s^{01} and r_p^{01} use the same n_f . Birefringence is not modelled: the fit reports only a difference of effective indices Δn_{eff} (see below).

2.3 Airy summation

The total reflection from the film is the sum of all the multiple internal re-reflections, which converges as a geometric series:

$$R_\sigma^{Airy} = \frac{r_\sigma^{01} + r_\sigma^{12} e^{2i\phi_\sigma}}{1 + r_\sigma^{01} r_\sigma^{12} e^{2i\phi_\sigma}}, \quad \sigma \in \{s, p\}$$

The phase for a single pass through the film is:

$$\phi_s = \frac{2\pi n_f d \cos \theta_1}{\lambda}, \quad \phi_p = \frac{2\pi n_p d \cos \theta_{1,p}}{\lambda}$$

For an isotropic film (the default, $\Delta n = 0$) $n_p = n_f$ and $\phi_s = \phi_p$.

2.4 Polarization collapse and unpolarized combination

The source (Perkin Elmer Lambda 950) provides no controlled, quantifiable polarization, and with unpolarized illumination the polarization degrees of freedom are not identifiable from the data. The current model therefore **collapses** them (`_COLLAPSE_POL`): the s/p weight is fixed to the unpolarized value $pol = 0.5$, the uniaxial birefringence and the s/p coherence phase are set to zero, and the measured reflectance is the equal-weight average of the two polarizations:

$$R_{\text{coh}}(\lambda) = \left[\frac{1}{2} |R_s^{\text{Airy}}|^2 + \frac{1}{2} |R_p^{\text{Airy}}|^2 \right] \cdot 100$$

2.5 Empirical fringe-contrast envelope

The measured fringe visibility is reduced by effects we do not control or quantify (imperfect source polarization, wavelength-dependent goniometer optics, partial coherence). These are lumped into a smooth envelope $V(\lambda)$ that scales **only** the oscillating part of the spectrum, leaving the fringe-free mean unchanged:

$$R(\lambda) = R_{\text{mean}} + V(\lambda) [R_{\text{coh}}(\lambda) - R_{\text{mean}}] f_{\text{inh}}$$

where R_{mean} is the interface-only (fringe-free) reflectance obtained with $r^{12} \rightarrow 0$, and f_{inh} is the inhomogeneity damping (§4). The envelope is a vertex-form parabola with a directly controllable vertex wavelength λ_v :

$$V(\lambda) = f_{c0} + f_{cc} \left(\frac{\lambda - \lambda_v}{1000} \right)^2, \quad \lambda_v = 1500 + 1000 f_{cv}$$

clipped to $[0, 3]$. f_{c0} is the overall level, λ_v the vertex wavelength ($f_{cv} \in [-2, 2] \rightarrow \lambda_v \in [-500, 3500]$ nm), and f_{cc} the curvature/intensity. Placing λ_v outside the measured band makes V **monotonic** over the band (increasing or decreasing); $f_{cc} = 0$ gives a constant $V = f_{c0}$ (the default).

Why — The three sliders *pol/dn/pm* carry $f_{c0}/f_{cv}/f_{cc}$. The term is declared **empirical**: it scales amplitude only and never shifts the fringe positions (set by $n \cdot d \cdot \cos\theta$ in the Airy phase), so it cannot bias the recovered n, k, d . A contrast extremum inside the band would have no physical justification, hence the vertex is pushed out of band; in practice a constant level ($f_{cc} = 0$) is used, with the vertex/curvature reserved for testing.

What can go wrong at 60°, and the fix. With a nonzero curvature f_{cc} the envelope $V(\lambda)$ is a parabola, and a symmetric parabola that deepens the fringes in the visible *also* lifts them on the far-IR side. The lowest-order far-IR fringe can then come out with a flat, **trapezoidal** top instead of a rounded one. This is purely a distortion of the fringe *amplitude* by the envelope, not the sample: the fringe **positions** (which set n, k, d) are untouched. To recover a rounded top, move the vertex λ_v out of the measured band so the envelope is monotonic (no in-band curvature) and turn **MONO** on.

2.6 Effective-index difference

PSi with medium porosity has columnar morphology (columns $\perp$ the surface) and is form-anisotropic, so it **is** birefringent — and when it is, the birefringence **shows in the data**, through the angle-dependence of the effective index and, robustly, through its sign. What unpolarized reflectance does **not** give is the birefringence as a direct *value* $n_e - n_o$: the extraordinary index n_e cannot be separated (that would need a polarizer), and the fit returns at each angle a single **effective** index $n(\theta)$. We therefore report the clear, unambiguous quantity — the difference of the effective indices between the oblique-angle and the near-normal fit, over the transparency window ($k \approx 0$):

$$\Delta n_{\text{eff}}(\theta) = n(\theta) - n(8 \text{ deg})$$

Δn_{eff} is defined purely by this difference and must **not** be read as the birefringence $n_e - n_o$, of which we have no direct *measurement*. It is a mixed, angle-resolved quantity (it also folds in the geometry of the oblique measurement) and is consistent with the single-effective-index fringe relation of the published method (Mula 2026). Its **sign** is a robust, reproducible feature: on PSi(100) it comes out negative, in line with the published finding that the PSi(100) birefringence is negative — a morphological fingerprint (pore shape/distribution/connectivity) that sphere-based Bruggeman/EMA predicts with the wrong sign.

2.7 Angular spread (HOBL configuration)

At 40 deg and 60 deg the Lambda 950 incident beam has a non-negligible angular spread due to the illumination geometry. The code models this as a **three-angle average** equispaced around the nominal, with weights (w_c, w_l, w_r) and half-aperture da :

$$R(\lambda) = w_c \cdot R(\lambda; \theta_0) + w_l \cdot R(\lambda; \theta_0 - da) + w_r \cdot R(\lambda; \theta_0 + da)$$

The weights are normalized to $w_c + w_l + w_r = 1$ (with a safety clamp 10^{-9} to avoid divisions by zero). Implemented in `physics.calc_refl_3angle`. The parameters w_c , w_l , w_r , da are optimized as part of phase 2 of `run_fit` on the HOBL angles.

3 $n(\lambda)$ and $k(\lambda)$ model

3.1 Base nodes and extended nodes

The film is parametrized by **14 base nodes** in wavelength, at nominal positions:

```
nodes_wl_base = [240, 270, 300, 335, 375, 430, 470, 520, 650, 900, 1100, 1583, 2067, 2550] nm
```

To improve the numerical conditioning of the fit, each interval between two base nodes is split into 3 uniform sub-intervals, inserting 2 sub-nodes. This yields about **40 extended nodes** (*ext_nodes*) used by the optimization algorithm. The construction is in `physics._build_ext_nodes`:

$$\text{ext_nodes} = \text{sort}\left(\text{nodes} \cup \left\{ a + \frac{b-a}{3}, a + \frac{2(b-a)}{3} \right\}_{(a,b)}\right)$$

Of the 14 base nodes, 5 have a mobile position during PSi fits (sliders W300, W335, W375, W400, W1100 in D1). The other nine are fixed. For known films (SiO_2 , TiO_2 , ...) the 5 mobile ones are also locked to the nominal values.

3.2 PCHIP interpolation

The continuous $n(\lambda)$ and $k(\lambda)$ curves are reconstructed from the node values via **Piecewise Cubic Hermite Interpolating Polynomial** (PCHIP, `scipy.interpolate.PchipInterpolator`). PCHIP is locally monotone: the interpolant does not overshoot between monotonically increasing/decreasing nodes. This is desirable for n and k which, physically, should have controlled smoothness without spurious oscillations induced by the interpolation.

Final result physically clipped: $1 \leq n \leq 12$, $0 \leq k \leq 15$.

3.3 Nonlinear slider maps

To give high resolution in the regimes of interest, the internal sliders use nonlinear scales mapped to the physical values via:

$$n = 1 + \frac{x^3}{80}, \quad x = (n - 1)^{1/3} \cdot 80^{1/3}$$

$$k = 10^x \quad \text{for } x \geq -4, \quad k = 0 \quad \text{otherwise}$$

The functions are in `physics.n_map / k_map / inv_n_map / inv_k_map`. The cube-root map for n expands the resolution near $n = 1$ (UV regime for PSi) compressing it at high values; the log map for k makes the sensitivity uniform over 4 decades.

4 Instrumental corrections

The pure Fresnel/Airy forward model does not reproduce the data for several instrumental reasons: surface roughness, lateral film inhomogeneity, volume scattering, the jump at the spectrophotometer detector change. The code introduces four physically motivated corrections that act **only on the fringe amplitude**, never on their spectral position. This is the **amplitude/position separation** principle.

4.1 Roughness and scattering: f_{scatt}

Damping from surface roughness + volume scattering, applied as an amplitude factor on the internal propagation factor $e^{2i\phi}$:

$$f_{\text{scatt}} = \exp\left[-\frac{\text{scatt}}{130} \cdot \left(\frac{500}{\lambda}\right)^{1.6} \cdot \frac{1}{\cos\theta_1}\right]$$

The form is inspired by a Debye-Waller term with an empirical scale. The exponent 1.6 on the spectral dependence is a simplification of the formula $(500/\lambda)^{0.8}$ squared. The division by $\cos\theta_1$ accounts for the physical path in the film. At $\theta = 0$ the value degenerates to $f_{\text{scatt}} = \exp(-(\text{scatt}/130) (500/\lambda)^{1.6})$.

4.2 Lateral inhomogeneity: f_{amp}

Lateral thickness variations (beam footprint) produce an average of fringes at different phases which damps the oscillations. Defining $\sigma_d = d \cdot (\text{inhom}/100)$ the standard variation of d :

$$\phi_{\text{norm}} = \frac{\pi n_f \sigma_d \cos\theta_1}{\lambda}, \quad \phi_{\text{foot}} = \frac{\pi n_f \sigma_d (1/\cos\theta_0 - 1)}{\lambda}$$

$$f_{\text{amp}} = \exp[-2(\phi_{\text{norm}}^2 + \phi_{\text{foot}}^2)]$$

ϕ_{norm} is the standard contribution (phase variation per variation of d). ϕ_{foot} is the additional geometric contribution due to oblique illumination: at $\theta = 0$ it vanishes, at 60 deg it grows and contributes comparably to the normal one ($1/\cos 60 \text{ deg} - 1 = 1$, so $\phi_{\text{foot}} \approx \phi_{\text{norm}}$).

The amplitude factor is applied as an oscillation around the mean reflectance (interface-only Fresnel, no interference). Combined with the empirical contrast envelope $V(\lambda)$ of §2.5 (f_{amp} is exactly the f_{inh} factor there):

$$R_{\text{coh}, d} = R_{\text{mean}} + V(\lambda) (R_{\text{coh}} - R_{\text{mean}}) f_{\text{amp}}, \quad R_{\text{mean}} = \frac{1}{2} (|r_s^{01}|^2 + |r_p^{01}|^2) \cdot 100$$

Why — The f_{amp} term damps only the oscillations around the mean, without shifting the positions of peaks and valleys. This preserves the robustness of the thickness fit (which depends on the maxima positions) even in the presence of lateral inhomogeneity: scattering and inhomogeneity are **strongly degenerate with each other** but **not confusable with $n \cdot d$** — one of the method's strengths.

4.3 Global Debye-Waller: f_{DW}

An additional Debye-Waller term is applied to the whole $R(\lambda)$ (not only the oscillations), to model the damping of the Si substrate peaks too:

$$f_{\text{DW}} = \exp\left[-0.013 \cdot \frac{\text{scatt}}{130} \cdot \left(\frac{500}{\lambda}\right)^2 \cdot \frac{1}{\max(\cos^2\theta_0, 0.1)}\right]$$

Typical effect: < 2% at 8 deg, up to ~15% at 60 deg in the UV with $\text{scatt} \approx 100$. The constant 0.013 is an empirical scale factor.

4.4 IR correction at the detector change (lambda > 890 nm)

The Lambda 950 changes detector (PMT → PbS) at 890 nm, introducing a small multiplicative jump and a non-flatness of the IR instrumental response. For $\lambda > 890$ nm (and only for the angles admitted by the code's IR configuration):

$$R(\lambda) \rightarrow R(\lambda) \cdot kj \cdot \left(\frac{890}{\lambda}\right)^\alpha \cdot \exp\left[-\beta \frac{\lambda - 890}{1000}\right] \quad (\lambda > 890 \text{ nm})$$

The three parameters kj (slider IR Jump, default 1.0), α (slider IR Exp), β (slider IR Dmp) are fitted. The multiplicative jump kj is separated from the spectral curvature (power α) and from the exponential damping (β), so that each captures a different feature of the detector transition.

4.5 Measuring the IR jump from the data (MEAS JUMP)

As an alternative to fitting kj , the MEAS JUMP button (D1) **measures it directly from the data** of the current angle as the ratio between the mean level just above and just below 890 nm (local polynomial fits on the two sides of the jump, with a small gap):

$$kj_{\text{meas}} = \frac{\bar{R}_{\lambda > 890}}{\bar{R}_{\lambda < 890}}$$

The value is set on the IR Jump slider and **locked**. Anchored to the data, it prevents the fit from confusing the instrumental jump with n , k or scattering. The measured ratio kj varies per sample and per angle: it moves further below 1 from 8 deg to 60 deg, i.e. the instrumental jump **grows** with the incidence angle (a larger detector-transition step at higher incidence), not the reverse.

4.6 Additive offset

A constant offset $offs$ (slider Offs, default 1.0%) compensates residual dark current and instrumental baseline. Applied after f_{DW} :

$$R(\lambda) = R_{\text{coh,d}}(\lambda) \cdot f_{\text{DW}}(\lambda) + offs$$

5 Fit objective function

The optimization minimizes a functional χ^2_{tot} that combines data residuals and physical/numerical constraint terms. It is implemented in the closure `obj(p)` inside `fit_engine.run_fit`, with variations in `run_fit_multi`.

A note on the symbol χ^2 . We keep the name by analogy: the functional plays the role a chi-square plays in a fit, and every term carries a **subscript** naming its operational role (χ^2_{data} , χ^2_{conv} , ...). It is **not** a statistical chi-square: the spectra carry no per-point σ , so there is nothing to normalize by. What the interface reports is accordingly labelled **fit obj** (the full penalized objective) and **RMS resid** (the data misfit, in %R), never « χ^2 ».

5.1 Main term: weighted residuals

$$\chi^2_{\text{data}} = \sum_{\lambda} w(\lambda) \cdot [R(\lambda) - R_{\text{exp}}(\lambda)]^2$$

The spectral weighting $w(\lambda)$ is **uniform** ($w = 1$) for the near-normal configurations ($\theta \in \{8 \text{ deg}, 20 \text{ deg}\}$). For the HOBL configurations ($\theta \in \{40 \text{ deg}, 60 \text{ deg}\}$) a non-uniform weighting is applied that emphasizes the regime of the dominant fringes:

```
HOBL weighting (40 deg, 60 deg):
w(lambda) = 0 if lambda < lmin
w(lambda) = 15 if 400 <= lambda < 600
w(lambda) = 40 if 600 <= lambda <= 1300
w(lambda) = 0.5 if lambda > 1500
w(lambda) = 1 elsewhere
Plus extra UV: chi2 += 40 * sum((R-Rexp)^2) for lambda < min(500, lmin)
```

5.2 Optical-thickness anchoring term (HOBL only)

For HOBL fits, a very strong term penalizes deviations of the mean optical thickness $\langle n \rangle \cdot d$ from the reference value determined at the start of the fit:

$$\Delta = \frac{\langle n_p \rangle \cdot d - \langle n \rangle_0 \cdot d_0}{\langle n \rangle_0 \cdot d_0}$$

$$\chi^2_{\text{anchor}} = 10^9 \cdot \Delta^2 \text{ if } |\Delta| > 0.01, \quad 0 \text{ otherwise}$$

The reference value $\langle n \rangle_0 \cdot d_0$ is computed at the start of `run_fit` from the initial values on the extended nodes (`opt_thick_ref = np.mean(n_ext0) * d_val`). The 1% threshold is the dead zone where the anchor is off; beyond it, the weight 10^9 makes the deviation practically impossible.

Why — The optical-thickness anchor blocks the $n \times d$ **degeneracy**, which is universal for any single-wavelength method: moving n and d jointly in inverse proportion does not change the fringe positions. The term forces the HOBL fit to stay close to the initial solution of the near-normal fit (8 deg/20 deg), which has already resolved the degeneracy via geometric triangulation. When only one angle is available — no multi-angle reflectance, likely the commonest case — the two-angle triangulation is not an option: there the degeneracy is broken instead by the semi-automatic **d-scan** (§9.5, and the procedure in D3), which brackets d from curated fits at several thicknesses, and that bracket anchors the fit.

5.3 Regularization (smoothness)

Quadratic penalty on the second derivative of the nodes:

$$\chi^2_{\text{reg}} = w_{\text{reg},n} \sum_j [\nabla^2 n_j]^2 + w_{\text{reg},k} \sum_j [\nabla^2 k_j]^2$$

The weights are **different between HOBL and near-normal** by choice:

near-normal (8 deg, 20 deg): $w_{\text{reg}_n} = 100$, $w_{\text{reg}_k} = 50$, $mw = 1e6$
 HOBL (40 deg, 60 deg, sing.): $w_{\text{reg}_n} = 50$, $w_{\text{reg}_k} = 30$, $mw = 1e9$

5.4 Monotonicity ($\lambda \geq 420$ nm, for both n and k)

When the user enables MONO n / MONO k (toggle in D1), a quadratic penalty is added that becomes active only if the first difference changes sign in the forbidden direction:

$$\chi_{\text{mono},n}^2 = m_w \sum_{j \in M_n} \max(\Delta n_j, 0)^2$$

$$\chi_{\text{mono},k}^2 = m_w \sum_{j \in M_k} \max(\Delta k_j, 0)^2$$

where $\Delta n_j = n_{j+1} - n_j$ and analogously for k . The masks M_n and M_k are:

- M_n : ext_nodes with $420 \leq \lambda \leq 2600$ nm. Excludes the UV region ($\lambda < 420$) where PSi shows **anomalous dispersion** near the Si interband transitions.
- M_k : the same mask, ext_nodes with $420 \leq \lambda \leq 2600$ nm. k starts rising at shorter wavelengths than n , so the same 420 nm boundary is safe for both.

5.5 Convexity of n and k

Over 420–2500 nm both n and k are also pushed to be **convex**: a negative second difference is penalized. Combined with the monotonic decrease this is the shape both quantities physically have — n follows normal dispersion (Cauchy, $n = A + B/\lambda^2$, so $d^2n/d\lambda^2 > 0$) and k is an absorption tail that decays and flattens towards $k \approx 0$ in the IR:

$$\chi_{\text{conv}}^2 = m_w \sum_j \max(-\nabla^2 x_j, 0)^2, \quad x = n, k$$

Convexity acts on **both** n and k at **every** angle: it is a property of the material's dispersion, not of the incidence angle. Each is gated on its own toggle: switching **MONO n** (or **MONO k**) off removes *all* of that quantity's shape priors — monotonicity, convexity and anti-zigzag. A user lock wins over the fit's own constraints.

5.6 Anti-zigzag (both n and k)

Penalizes sign changes of the first derivative (spurious oscillations between adjacent nodes):

$$\chi_{\text{zz},n}^2 = m_w \sum_j \max(-(\Delta n_j)(\Delta n_{j+1}), 0)^2$$

Analogous for k . The quantity $(\Delta n_j)(\Delta n_{j+1})$ is negative when the sign of the derivative changes between two consecutive intervals; in that case the penalty activates.

5.7 IR maxima position anchoring (40 deg and 60 deg)

At the oblique angles the IR fringes are **wide** (small curvature): the smoothness/derivative penalty has a positional grip proportional to the peak curvature, so on wide fringes it barely constrains the position — the maximum center can slide by tens of nm even with correct amplitude. A term is then added that anchors **the position directly** for each IR maximum via a *smooth weighted centroid* (local linear detrend, weight g on the upper half of the peak):

$$\lambda_c = \frac{\sum_i \lambda_i g_i}{\sum_i g_i}, \quad g_i = \max(0, R_i - R_i^{\text{trend}})^2$$

$$\chi_{\text{pos}}^2 = w_{\text{pos}} \sum_p (\lambda_c^{(p)} - \lambda_{\text{target}}^{(p)})^2$$

The positional stiffness is $\propto w_{\text{pos}} \cdot \delta$ and is **independent** of the peak curvature (unlike the derivative penalty). The term is C^1 (smooth for L-BFGS-B).

Gate: active at **40 deg and 60 deg**, only for maxima at $\lambda \geq 1000$ nm; at 60 deg it uses a stronger weight (*peak_pos_w_hi*). In the collapsed model the 60 deg maxima are **shifted** (not deformed), so the position term applies there too.

5.8 Optional maxima-centering aid (FRINGE AID)

For samples with **few, low-contrast fringes**, an optional term helps center the modelled maxima on the measured ones: a derivative/concavity penalty that pushes R toward first derivative ≈ 0 (weight 5×10^4) and concavity ≤ 0 (weight 10^8) at the measured peak positions. It is **opt-in** (default OFF, toggle AID in D1) and, when on, acts only for $\lambda \geq \text{peak_pen_lam_min}$ (500 nm), skipping the deep blue.

5.9 Total functional

To the functional below are added, when applicable, the χ_{pos}^2 position term (40 deg / 60 deg) and the opt-in maxima-centering term (FRINGE AID) just described.

$$\chi_{\text{tot}}^2 = \chi_{\text{data}}^2 + \chi_{\text{anchor}}^2 + \chi_{\text{reg}}^2 + \chi_{\text{mono},n}^2 + \chi_{\text{mono},k}^2 + \chi_{\text{conc},k}^2 + \chi_{\text{zz},n}^2 + \chi_{\text{zz},k}^2$$

χ_{tot}^2 above is the **optimizer objective** (weighted SSE plus the constraint terms) — a least-squares cost, not a statistical χ^2 . The **fit-quality** number reported in the UI, in the .par header and in the d-scan window is a different quantity: the **RMS reflectance residual**, $\text{RMS} = \sqrt{\text{mean}[(R - R_{\text{exp}})^2]}$ in %R units, from the data residuals only (no constraint terms, no spectral weighting). The spectra carry no calibrated per-point σ , so this is an unweighted least-squares misfit — **not** a σ -normalized (statistical) reduced χ^2 ; its scale reflects the residual amplitude directly, not statistical consistency. Statistical validation goes through region-wise KK + triangulation of d , not a noise-normalized χ_r^2 .

6 Optimization

6.1 L-BFGS-B algorithm

All fits use `scipy.optimize.minimize` with the 'L-BFGS-B' method (Limited-memory BFGS with bound constraints). Stop parameters (typical configurations):

```
near-normal (8 deg, 20 deg): maxiter=5000, ftol=1e-18, gtol=1e-12, eps=1e-8
HOBL (40 deg, 60 deg): maxiter=2000, ftol=1e-12, gtol=1e-7, eps=1e-6
```

The bounds for each parameter are built in the setup of `run_fit`: for d , $\pm 0.5\%$ is used (known films: $\pm 50\%$), for $scatt$ $\pm 2\%$ of the current value, for each node (n, k) a wavelength-dependent bound via `physics._node_bounds`:

```
lambda <= 375 -> n in (1, 5), k in (0, 2.5)
lambda <= 470 -> n in (1, 4), k in (0, 1.5)
lambda <= 520 -> n in (1, 4), k in (0, 0.20)
lambda <= 650 -> n in (1, 4), k in (0, 0.05)
lambda <= 900 -> n in (1, 4), k in (0, 0.010)
lambda <= 1300 -> ...
etc. (see physics._node_bounds)
```

6.2 Lock system

The state keeps four lock lists:

- `state.lock_left` (13 elements): the left-side parameter sliders.
- `state.lock_n` (14): one lock per n base node.
- `state.lock_k` (14): one lock per k base node.
- `state.lock_extra` (3): [dn, pm, ps1].

When a slider is locked, the setup builds bounds of the form (v, v) (null range) on the corresponding parameter: the algorithm leaves the value unchanged. For the n, k nodes, the lock is applied via `physics._apply_lock_nk_bounds` which, besides fixing the locked base nodes, also fixes the sub-nodes between two adjacent base nodes that are both locked (for visual consistency: if the user locks both ends of an interval they expect the PCHIP interpolation of that stretch to be fixed too).

6.3 Multi-stage in `run_fit_multi`

The `run_fit_multi` function performs the global constrained-consistency fit over multiple angles, in three stages:

- **Stage 1** (`obj_s1`): minimizes an aggregate functional that **imposes the same nodes** (n, k) for all angles, while the instrumental parameters (km, ks, kb) and the collapsed contrast-envelope params ($pol, dn, pm = fc_o, fc_v, fc_c$, §2.4–2.5) are **free per angle**. Angular weights: $w = 3$ at 8 deg, 2 at 20 deg, 1 at 40 deg, 1 at 60 deg (the low angle is considered the most reliable for n, k).
- **Stage 2** (`obj_s2`): refines the nodes per angle with tight bounds ($\pm 5\%$ relative to stage 1) to reabsorb small per-angle residuals without one dragging the others.
- **Stage 3** (multi-start, only if HOBL present): a multi-start on the per-angle pm parameter (6 values in $[-\pi, \pi]$), re-running stage 2 from each, to escape local minima. In the collapsed model pm is the contrast curvature fc_c .

After the three stages, a single `run_fit` is run for each angle as the final refinement.

7 Uncertainty estimation

7.1 Numerical diagonal Hessian

For each node j of n and k , the uncertainty sigma is estimated by perturbing the single node by $\pm \epsilon$ ($\epsilon = 10^{-4}$) and computing the local curvature of the residual sum of squares:

$$f(x) = \sum_{\lambda} [R(\lambda; x) - R_{\text{exp}}(\lambda)]^2$$
$$\frac{\partial^2 f}{\partial x_j^2} \approx \frac{f(x + \epsilon e_j) - 2f(x) + f(x - \epsilon e_j)}{\epsilon^2}$$
$$\sigma_j = \frac{1}{\sqrt{\partial^2 f / \partial x_j^2}}$$

Implemented in `fit_engine._compute_sigma_nodes`. The σ is computed both for the 40 ext-nodes n and for the 40 k ; on saving (D1, .par and _data.csv) it is interpolated onto the 14 base nodes (for .par) and onto `wl_exp` (for _data.csv).

7.2 Limitations

The estimate is indicative rather than rigorous: it uses only the diagonal of the Hessian and neglects the parameter covariances, and its scale is taken from the residual sum of squares itself, which is circular for a proper variance-based error estimate and does not incorporate a separate instrumental-noise. It is reported as an indication of the relative node-to-node confidence, not as a rigorous confidence interval, a point returned to in the discussion of reliability.

7.3 Propagation sigma -> sigma_n_KK via Hilbert

The subtractive integral is linear in $\epsilon_2 = 2nk$, so its uncertainty propagates analytically — but what it returns is ϵ_1 , **not** n . Propagation therefore takes two steps. Defining $\sigma_{\epsilon_2} = 2n\sigma_k$ (neglecting $\sigma_n \ll \sigma_k$), the first gives σ of ϵ_1 through the **same subtracted kernel** the MSKK-1 transform uses — the difference of the two poles, at E_0 and at the IR anchor:

$$\sigma_{\epsilon_1}^2(\lambda_0) = \left(\frac{2\Delta E}{\pi}\right)^2 \sum_j [(K(E_j, E_0) - K(E_j, E_{IR})) \cdot \sigma_{\epsilon_2}(E_j)]^2$$

$$K(E_j, E_0) = \frac{E_j}{E_j^2 - E_0^2}, \quad (\text{excluding points with } |E_j - E_0| < 2.5 \Delta E)$$

The second step carries σ_{ϵ_1} onto n through the Jacobian of the inversion $n = \sqrt{((\epsilon_1 + |\epsilon|)/2)}$ used above. In the transparency window ($\epsilon_2 \rightarrow 0$) it reduces to $1/(2n)$; it departs from that where the film absorbs:

$$\sigma_{n_{KK}}(\lambda_0) = \sigma_{\epsilon_1}(\lambda_0) \cdot \frac{\partial n}{\partial \epsilon_1}, \quad \frac{\partial n}{\partial \epsilon_1} = \frac{1}{4n} \left(1 + \frac{\epsilon_1}{|\epsilon|}\right) \rightarrow \frac{1}{2n} \quad (\epsilon_2 \rightarrow 0)$$

Implemented in `fit_engine.calcola_kk_uncertainty`. The result is displayed as a band in the KK window and saved as the column `sigma_KK_propagated` in `_kk_grid.csv`.

What this σ is, and what it is not. It is the uncertainty the **transform** adds given σ_k , and it goes to zero at the IR anchor: there ϵ_1 is imposed by the fit rather than predicted, so the pinch in the band marks the one wavelength that is not a KK result. The anchor's own uncertainty is deliberately excluded — it is built from n_{fit} , hence common to n_{fit} and n_{KK} , and adding it in quadrature would inflate the very error bar used to judge $|n_{\text{fit}} - n_{KK}|$. Sizing it properly would need the covariance between the two, which this indicative estimate does not carry: as everywhere else here, the σ bounds the numerics, not the physics — whether the two curves ought to agree is the operator's call.

8 Kramers-Kronig relation

To validate the physicality of the n, k extracted from the fit, the code computes $n_{KK}(\lambda)$ from the Hilbert transform of $k(\lambda)$ and compares it point-by-point with the fit's $n(\lambda)$. The $\text{RMS}(n - n_{KK})$ metric is the main physical-consistency output.

8.1 Adopted Kramers-Kronig form: MSKK-1

The code uses the **Multiply Subtractive Kramers-Kronig** with one subtraction (MSKK-1), in the $\epsilon_1(E)$ formulation:

$$\epsilon_1(E_0) = \epsilon_1(E_{\text{IR}}) + \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{E' \epsilon_2(E')}{E'^2 - E_0^2} dE' - \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{E' \epsilon_2(E')}{E'^2 - E_{\text{IR}}^2} dE'$$

where $E_{\text{IR}} = 1240/2500$ is the IR anchor. Subtracting the constant $\epsilon_1(E_{\text{IR}})$ cancels the unknown contribution of the high-energy states that fall outside the computation range (UV-VUV); this makes the result relative to an anchor measurable in the IR (where PSi is transparent and $n(2500)$ is well known).

Implemented in `fit_engine.calcola_kk` (integral on an equispaced grid in E , 35000 points between 0.05 eV and 30 eV, output on `wl_out` at 2 nm step).

8.2 Cauchy singularity

The Hilbert integral has a pole at $E' = E_0$. The code handles the singularity with a **punctum excisum**: the grid points with $|E' - E_0| < 2.5 \Delta E$ are excluded from the sum:

$$\text{kernel}_{\text{reg}}(E_j, E_0) = \frac{E_j}{E_j^2 - E_0^2} \text{ if } |E_j - E_0| \geq 2.5 \Delta E, \quad 0 \text{ otherwise}$$

A strategy equivalent to regularizing the principal value by removing a small symmetric window around the singularity: the introduced error is of order $\epsilon_2(E_0) \cdot (2.5 \Delta E)^2 / E_0$, negligible for E_0 in the visible range.

8.3 Boundary artifact at 2550 nm

The truncation of the integral at $E_{\text{min}} \approx 0.05$ eV (corresponding to the maximum λ in the computation grid) introduces a boundary artifact at the 2550 nm node: $n_{KK}(2550)$ tends to drift downward iteration after iteration of the fit-KK loop.

Why — The standard truncation corrections (asymptotic extension, tail fit) are hard to apply to PSi because the behavior of $k(\lambda)$ above 2.5 micron is poorly known. The code does not correct the artifact analytically; it documents that the 2550 nm node should be locked manually in the 4-step KK protocol (D1 Sec. 5) when it drifts below 95% of the previous cycle's value.

8.4 Savitzky-Golay smoothing

The raw $n_{KK}(\lambda)$ result has numerical noise due to the discretization and the regularized singularity. The smoothing applied is Savitzky-Golay with `polyorder = 3` and a user-controlled window (Smoothing window slider in D1, odd values, default 1 = no smoothing). Implemented with `scipy.signal.savgol_filter`.

8.5 Consistency metric

Four metrics computed in the KK window (in D1):

$$\text{RMS}(n - n_{KK}) = \sqrt{\frac{1}{N} \sum_{j \in M_\lambda} [n(\lambda_j) - n_{KK}(\lambda_j)]^2}$$
$$\max|\Delta n/n| = \max_{j \in M_\lambda} \frac{|n(\lambda_j) - n_{KK}(\lambda_j)|}{n(\lambda_j)}$$

The mask M_λ is $[\lambda_{\min}, 2550 \text{ nm}]$, where $\lambda_{\min}$ is the value of the $\lambda_{\min}$ RMS slider (default 420 nm).

The $\sigma_{KK}(\text{local})$ is the local standard deviation of $n_{KK}(\text{raw})$ over a sliding window of 15 points: it measures the intrinsic noise of the KK computation, independently of the smoothing. The RMS/σ_{KK} ratio indicates whether the $n - n_{KK}$ disagreement is explained by the computation noise (values 2–3) or is a real physical discrepancy (values $\gg 3$).

9 Thickness triangulation (Bragg/Snell)

The reflectance-maxima method, implemented in the Peak Analysis Window (D1 Sec. 6), allows estimating the thickness d in a **model-independent** way: it uses only the spectral position of the maxima at two different angles, without assuming a functional form of $n(\lambda)$.

9.1 Bragg condition for the maxima

A reflectance maximum occurs when the total phase of the double pass in the film is an integer multiple of 2π :

$$2n_f(\lambda) d \cos \theta_1 = m \lambda, \quad m = 1, 2, 3, \dots$$

With Snell: $\cos \theta_1 = \sqrt{(1 - \sin^2 \theta_0) / n_f^2}$. The order m is that of the maximum ($m = 1$ for the first from IR toward UV, $m = 2$ for the second, etc.).

9.2 Dispersion-free two-angle solution

Consider a maximum of order m observed at two angles θ_A and θ_B , at wavelengths λ_A and λ_B . We impose the identity of m (same order):

$$\frac{2n_f(\lambda_A) d \cos \theta_{1,A}}{\lambda_A} = \frac{2n_f(\lambda_B) d \cos \theta_{1,B}}{\lambda_B}$$

If $\lambda_A \approx \lambda_B$ (nearby angles), the dispersion $n_f(\lambda)$ is approximately equal at the two angles, and n_f is eliminated between the two equations:

$$\left(\frac{\lambda_A}{\lambda_B}\right)^2 = \frac{n_f^2 - \sin^2 \theta_A}{n_f^2 - \sin^2 \theta_B}$$

Solving for n_f :

$$n_f^2 = \frac{\sin^2 \theta_A - (\lambda_A/\lambda_B)^2 \sin^2 \theta_B}{1 - (\lambda_A/\lambda_B)^2}$$

Once n_f is known, substitute in Bragg to obtain the thickness:

$$d = \frac{m \cdot \lambda_A}{2n_f \cos \theta_{1,A}}, \quad \cos \theta_{1,A} = \sqrt{1 - \sin^2 \theta_A / n_f^2}$$

COMPUTE d (8/20) button in the Peak Window (it returns the thickness d only). Implemented for the 8 deg/20 deg pair (see D1 Sec. 6).

Why — The triangulation is **independent of the dispersive model** because, on the fine λ scale between the same maxima at 8 deg and 20 deg, $n_f(\lambda)$ is effectively constant. This makes d immune to errors in the n,k model: even if the fit gives a wrong n,k , the thickness extracted from the peaks stays correct. This is the methodological cornerstone of the thickness determination.

9.3 Effective index from single-angle maxima (n_{eff})

The same Bragg condition, once the thickness d is known, can be solved for n at each maximum (λ , m) of a single angle. With Snell $\theta_r = \arcsin(\sin \theta / n)$:

$$2 n d \sqrt{1 - \left(\frac{\sin \theta}{n}\right)^2} = m \lambda \Rightarrow n_{\text{eff}}(\lambda)$$

solved numerically (brentq in the physical range 1.1–3.5). It is the index shown in the **MAXIMA** view (D1). At oblique incidence n_{eff} is an **effective** quantity (mixed in polarization and, on PSi, possible anisotropy), not the ordinary index: this is why at 60 deg only n_{eff} is reported and not k .

The maxima positions fix only the product $n \cdot d$ (degeneracy: raising n and lowering d leaves the fringes unmoved). To extract n_{eff} an independent d is therefore needed — from the fit, from triangulation, or a custom value typed by the operator (e.g. the d_{eff} from the d-SCAN), which the Peak Window accepts and feeds into the

n_{eff} calculation.

9.4 Adaptive maxima detection

The expected spacing between consecutive maxima, from the Bragg condition, grows as λ^2 :

$$\Delta\lambda(\lambda) = \frac{\lambda^2}{2n(\lambda)d\cos\theta_r}$$

The peak detector uses an **adaptive** minimum distance equal to a fraction of $\Delta\lambda$, capped above by a fixed maximum distance s_{max} :

$$s(\lambda) = \min(f \cdot \Delta\lambda(\lambda), s_{\text{max}})$$

In the visible ($\Delta\lambda$ small) s drops below s_{max} and resolves the closely-spaced fringes; in the IR ($\Delta\lambda$ large) the cap fixes $s = s_{\text{max}}$, so the IR maxima — the ones that determine d — stay **identical** and the d estimate does not change.

9.5 Determining d as a bracket: RMS residual(d) and KK residual(d)

The triangulation is model-independent but **ill-conditioned** when the angular splitting of the maxima is small. A complementary, more constrained estimate is obtained by scanning d : for a few values of d the operator runs a careful fit and two quantities are observed.

(1) The **RMS reflectance residual on the data** as a function of d , with the (parabolic) minimum as the estimate (an unweighted least-squares misfit, not a σ -normalized reduced χ^2):

$$d^* = \text{argmin}_d \text{RMS}(d)$$

(2) The **KK-consistency residual** averaged in the transparency-edge region (where k is small but nonzero), cleaner than the RMS residual on non-ideal samples:

$$\rho_{KK}(d) = \langle |n_{\text{fit}} - n_{KK}| \rangle_{[480, 750] \text{ nm}}, \quad d^* = \text{argmin}_d \rho_{KK}(d)$$

The flat bottom of each curve around its optimum **indicates** a bracket rather than a point for d , with sharp rising branches away from it. That width is relative to the single indicator — **not a rigorous uncertainty on d** : no one metric bounds d by itself. The indicators of the d -scan together **suggest where d lies**; none, on its own, measures the confidence of that estimate.

10 Physical constraints and priors

Three shape/validity choices are built into the program. Two are physical facts of the problem — the monotonicity boundary at 420 nm and the restriction of KK to near-normal incidence — and should not be moved. The third, $k \approx 0$ in the transparency region, is a prior you can switch off (the **K-BOUND** toggle). The physical **why** of each is explained here.

10.1 Monotonicity of n only at $\lambda \geq 420$ nm

Crystalline silicon shows the E_1 and E_2 interband transitions around 367 nm and 295 nm respectively. In this region the real part of $\epsilon(\lambda)$ exhibits strong anomalous dispersion (n is not monotone): it rises approaching the k peak, then drops sharply above. PSi, being porous Si, inherits its optical structure with weight proportional to the *filling fraction* of the matrix.

Imposing monotonicity of n below 420 nm would force the fit to a physically wrong solution. This is why $M_n = \{\text{ext_nodes with } \lambda \geq 420 \text{ nm}\}$ in the monotonicity penalty term (Sec. 5.4).

Why — Extending the monotonicity below 420 nm would force the fit through the anomalous-dispersion region and destroy its ability to capture the E_1 interband, so the constraint is confined to $\lambda \geq 420$ nm.

10.2 k approximately 0 in the transparency region (K-BOUND)

PSi of medium porosity is essentially **transparent** in the red visible and the NIR, so above ~550 nm the fitted k is a calculation residue, not physical information. With the **K-BOUND** toggle on (the default), the fit clamps k there to ~0, uniformly across the single- and multi-angle paths:

```
K-BOUND on: lambda > 550 nm -> k bounds = (1e-9, 1e-6)
K-BOUND off: k follows the graded physical cap (_k_bounds / _node_bounds)
```

Why — A finite $k > 0$ in the transparency window would damp the fringes non-physically (degenerate with scatt), so clamping it is the right default for typical PSi (porosity 50–80%). It is a prior, not a law: for a very-low-porosity sample whose bandgap sits below 550 nm, or to check whether the data want a finite k , switch K-BOUND off and let it float. The clamp only shapes the free (PSi) fit; a known material keeps its tabulated k .

10.3 KK valid only at 8 deg

The traditional KK relation relates the real and imaginary parts of the dielectric function $\epsilon(\omega)$; applying it to reflectance introduces an approximate step that assumes **near-normal incidence** (θ_0 small, typically ≤ 10 deg). For larger angles, polarization and anisotropy (birefringence) effects violate the assumptions and the KK identity derived from R becomes a mere approximation.

The KK button in D1 (row 3 of the Main Window) is active only if the selected angle is 8 deg; for other angles the status-bar message is *KK: only valid at 8 deg*.

11 Symbol glossary

$n + i k$ — complex refractive index of the film. n real positive, k real ≥ 0 .

$\epsilon_1 + i \epsilon_2$ — complex dielectric function $= (n + i k)^2$.

d — physical film thickness in nm.

θ_0 — incidence angle in air (real, in degrees in the code).

θ_1, θ_2 — refraction angles in the film and in the substrate (complex, from Snell).

r_s, r_p — Fresnel coefficients for s and p polarization; superscript 01 for the air-film interface, 12 for film-substrate.

R^{Airy} — total film reflectance via Airy summation (geometric series of the multiple re-reflections).

ϕ_s, ϕ_p — phase of a single pass in the film for s and p polarization.

pol — s-polarized intensity fraction of the source, in $[0, 1]$. In the collapsed model fixed to 0.5 (unpolarized); the pol slider carries fc_0 , the contrast level (§2.4–2.5).

Δn — uniaxial birefringence $n_e - n_o$. Not accessible from unpolarized reflectance; the fit reports the effective-index difference Δn_{eff} (§2.6) instead.

ϕ_{mix}, ps_1 — s/p relative-phase parameters (goniometer, HOBL angles); set to 0 in the collapsed model.

$scatt$ — amplitude of the damping from roughness + volume scattering (Scatt. slider, default 5).

$inhom$ — lateral thickness inhomogeneity in %, $\sigma_d = d * inhom/100$.

$offs$ — additive offset of $R(\lambda)$ in % R.

km, ks, kb — parameters of the IR correction at $\lambda > 890$ nm (IR Jump, IR Exp, IR Damp).

dn, pm, ps_1 — in the collapsed model the dn/pm sliders carry the contrast-envelope params fc_v (vertex) / fc_c (curvature); ps_1 inactive.

wc, wl, wr, da — weights and half-aperture of the HOBL angular spread.

opt_thick_ref — reference value $\langle n \rangle * d$ at the start of the HOBL fit (optical-thickness anchor).

λ_{min} — lower λ threshold used for the residual weighting and for the RMS computation region of the KK window.

$w_{\text{reg}_n}, w_{\text{reg}_k}$ — smoothness regularization weights on n and k .

mw — single weight for monotonicity / anti-zigzag (10^6 in near-normal, 10^9 in HOBL).

E_{IR} — subtraction anchor of MSKK-1, fixed at 1240/2500 eV (corresponding to $\lambda = 2500$ nm).

$\sigma_{\text{KK_local}}$ — local standard deviation of raw n_{KK} , a measure of the intrinsic noise of the KK computation.

$\sigma_{\text{KK_propagated}}$ — n_{KK} uncertainty propagated from σ_k : through the subtracted Hilbert kernel to σ_{ϵ_1} , then onto n via $dn/d\epsilon_1$. It is what the transform adds, and it vanishes at the IR anchor (imposed, not predicted).